

Sparks of Surprise: Multi-objective Recommendations with Hierarchical Decision Transformers for Diversity, Novelty, and Serendipity

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ABSTRACT

Personalized Session-based Recommendation (PSR) extends the traditional sequential recommendation models—which typically recommends the next item based on a recent active session—to leverage historical sessions of a user for short-term recommendations in current session. However, existing PSR methods face two limitations: (1) treating offline sessions uniformly as static data and relying on user embeddings to represent personalized information overlook the dynamic evolution of interests over time, which can change significantly as sessions progress in practical application. (2) focusing on accuracy, i.e., recommending items relevant to recent interactions, ignores the balance of multi-faceted requirements for user satisfaction, i.e., diversity, novelty, and serendipity.

Therefore, we introduce Multi-objective PSR (MOPSR) task and propose Hierarchical Decision Transformers (HDT) framework, which models strictly sequential preference transitions of users across and within sessions to balance recommendation accuracy with the mentioned objectives. To address the first problem, Inter-session DT dynamically tracks the user’s long-term preference across sessions by maintaining a goal state. This goal state serves as personalized information to collaboratively make recommendations with short-term state via the Intra-session DT. To tackle the second limitation, we propose inter-session and intra-session unexpected returns to trade off relevant recommendations and user preferences on diversity, novelty, and serendipity. The hierarchical returns help the recommender accurately identify signals of the user’s expectations and changes in multi-objective preferences. To verify the effectiveness of our method on the MOPSR, we apply HDT to four state-of-the-art sequential recommendation models and conduct experiments on two publicly available datasets. Experimental results demonstrate that (1) HDT can widely generalize sequential models to solve the MOPSR task in scenarios with incrementally generated

sessions, and (2) our method can balance multi-objectives by maintaining and even enhancing accuracy while effectively improving the diversity, novelty, and serendipity objectives.

CCS CONCEPTS

• Information systems → Recommender systems.

KEYWORDS

Multi-Objective Personalized Session-based Recommendation, Reinforcement Learning, Diversity, Novelty, Serendipity

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1 INTRODUCTION

Recommender Systems (RS) help users discover content from large-scale data streams and promote engaging content [11, 37, 51] for users. In recent years, next-item recommendation systems have been at the forefront of research, e.g., in entertainment (e.g., video streaming) and e-commerce, with a strong focus on recommending the next relevant item based on user-item interactions within one recent active session. For the most part, sequential or next item recommendation methods [10, 15, 38, 52] attempt to model a short-term sequence of user-item interactions within one recent active session to predict the next recommendations for users. However, in many scenarios, recommendations are determined by both user’s long-term and short-term interests. For example, users may prefer revisiting past topics in music application [34] and forums [29]. Therefore, Personalized Session-based Recommendation [27, 29, 49, 53] is proposed to recommend the next item by considering both user’s past sessions and current session.

However, how to effectively model dynamic changes of user preference through historical sessions and short-term interest in the current session remains a challenging task. Some research treats all historical sessions as equivalent interactions and adapts the Graph



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Neural Network (GNN)-based methods [27, 38, 49, 53] extract either global or local graph relations from entire offline static data in advance to model items/users. However, they ignore the personalized information from the dynamic changes of user preference over time in a real-world scenario, where user session proceeds sequentially. Even user embeddings are trained to present personalized, they are ineffective due to insufficient training when user sessions are sparse. Consequently, GNN-based approaches fall outside the scope of our research. To be practical, we focus on PSR scenarios where user interactions are strictly sequential across and within sessions.

Moreover, Multi-objective PSR (MOPSR) that realizes trade-offs between accuracy, diversity, novelty, and serendipity remain an unsolved task. Existing PSR methods still align with most traditional recommendation models [10, 43, 45, 47] to optimize the metrics of accuracy, referring to recommend items that are relevant to recently interacted items, e.g., HR [10] and NDCG [13] metrics. However, a strictly relevance-oriented RS overlooks the intrinsic recommendation qualities, affecting user experience in terms of diversity (recommended items are dissimilar to the user’s consumption patterns), novelty (recommended items are new/unseen), and serendipity (recommendations are relevant to users and beyond expectations). Recently, a multi-objective recommender system [6] considering the diversity and novelty is proposed for the sequential recommendation by the offline reinforcement learning (RL) process. It trains multiple RL agents to balance accuracy with the two objectives in recommended items (the taken actions) by maximizing the expected cumulative multi-objective rewards (returns). However, traditional RL approaches, e.g., Q-learning [45, 46] cannot be directly applied to MOPSR since they tend to prioritize immediate rewards of action. This concentration on short-term gains in the current session poses a challenge in capturing multiple long-term preferences in past sessions. Furthermore, the training becomes complex due to setting different agent heads for each objective. Additionally, the limited offline user data cannot provide sufficient feedback for each agent, making it challenging to accurately tune and optimize for specific objectives.

To address the above issues, we introduce Multi-objective Personalized Session-based Recommendation (MOPSR) task to realize trade-offs between accuracy and diversity, novelty and serendipity objectives. First, the Hierarchical Decision Transformers (HDT) framework is proposed to personalize sequential models by concurrently modeling the user’s long-term preferences derived from historical sessions, and their short-term interests within the current session. Moreover, we propose inter-session and intra-session unexpected returns infused with HDT to capture the user’s signals of preferences on unexpected items to improve the balance between the mentioned objectives. Specifically, HDT contains two layers: Inter-session DT (InterDT) maintains user state transitions across sessions to generate goal states conditioned on the user’s future expectations; Intra-session DT (IntraDT) models state transitions by the user-item interactions within the current session, and the recommendation process is guided by the goal state from the InterDT with an extra intra-session novelty return.

The Hierarchical Decision Transformer (HDT) is trained on offline user data in a supervised manner, mitigating the limitations associated with Q-value estimation [45], Multi-agents [36], and high variances in traditional RL. During the inference, given the

user state from the previous session and the expectations we want to obtain for the next recommendations, HDT can directly take actions (recommend items) considering long-term and short-term objectives, e.g., serendipity and novelty. To verify the effectiveness of our approach, we apply HDT to four state-of-the-art sequential recommendation models and conduct experiments on two publicly available datasets. Experimental results are compared by specific objective metrics for accuracy, diversity, novelty and serendipity, which demonstrate the superior performance of our method in balancing those objectives.

Our contributions can be summarized as follows:

- We introduce Multi-objective Personalized Session-based Recommendation (MOPSR) task and propose a Hierarchical Decision Transformers (HDT) framework to recommend next items by leveraging user preference transitions across historical sessions.
- We infuse HDT with hierarchical unexpected returns in InterDT and IntraDT to comprehensively model long-term and short-term changes in user preferences for unexpected items.
- We establish specific metrics for multi-objectives to evaluate the trade-off capabilities of HDT. To validate the effectiveness of our method, HDT is applied to four state-of-the-art sequential recommendation models and experiments are conducted on two public datasets.
- The experimental results show that our method can balance multi-objectives by maintaining and even enhancing accuracy while effectively improving the diversity, novelty, serendipity.

2 METHODOLOGY

We first illustrate the Multi-objective Personalized Session-based Recommendation (MOPSR) task. Next, we introduce the overall framework of the proposed Hierarchical Decision Transformer (HDT) for MOPSR. Finally, we illustrate the modeling details.

2.1 Problem Formulation

• **Task Definition.** Different from Sequential Recommendation [15, 37, 38, 58, 59] or Session-based Recommendation [10, 18, 52] that recommend next items based on ordered/unordered interactions within a recent active session, Personalized Session-based Recommendation (PSR) [29] aims to recommend the next items based on users’ current session and historical sessions. PSR models [29] typically leverage the ordered information inside and across sessions to model the evolution of user preference for recommendations. Therefore, it is of great significance for platforms where user session data is incrementally generated. Let I denote the set of items, $C = \{C_1, \dots, C_h\}$ denotes a user’s sequence of sessions in chronological order, $C_i = \{x_{i,1}, \dots, x_{i,T}\} \in C (0 < i \leq h)$ denote the user-item interactions in each session, where $x_{i,j} \in I (0 < j \leq T)$ is the index of the interacted item ordered by a timestamp. Given historical sessions $C = \{C_1, \dots, C_{c-1}\}$ and the current session $C_c = \{x_{c,1}, \dots, x_{c,t}\}$, the goal is to recommend next item $x_{c,t+1}$ from the whole item set I that the user might be interested in.

• **Trade-off of Multi-Objectives.** We extend the PSR to the Multi-objective Personalized Session-based Recommendation (MOPSR) to balance recommendation accuracy with multiple objectives that affect user experience. First, general RSs aim to recommend the most relevant items to historical sessions that are often influenced

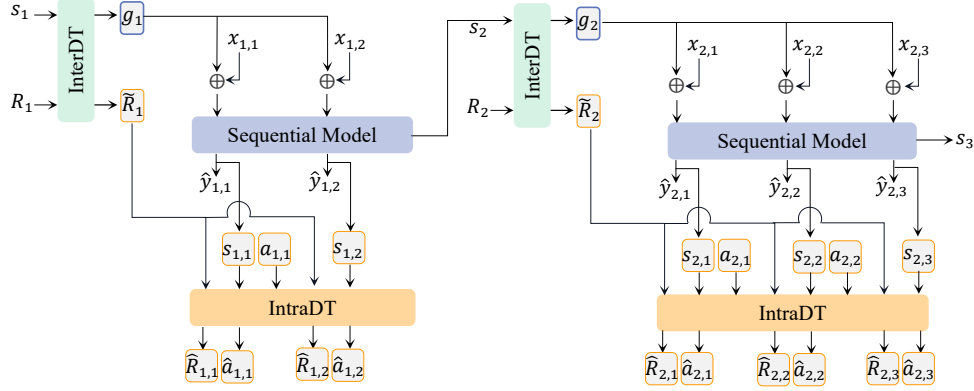


Figure 1: Example of personalizing sequential models by Hierarchical Decision Transformers (HDT). The InterDT observes the inter-session state s_c and inter-session unexpected return R_c , where $c \in \{1, 2, \dots\}$, to produce the goal state g_c and goal return $\tilde{R}_c$ for session C_c . s_1 is initialized by zeros. The IntraDT observes the intra-session state $s_{c,t}$, goal state g_c , goal return $\tilde{R}_c$, where $t \in \{1, 2, 3, \dots\}$, to predict the intra-session unexpected return $R_{c,t}$ and the action a_t in session c .

by popularity, leading to recommendations that are frequently seen while neglecting long-tail (new and unknown) ones. This reduces diversity and novelty [25, 36] in recommendations. Second, recommending relevant and unexpected items can provide users with serendipity [31]. However, considering past historical interests may bias recommendations towards items the already been seen or similar items to a user’s historical interactions for a PSR model, so that reduces unexpectedness. Conversely, overly recommending dissimilar items can reduce relevance. Therefore, the goal of our MOPSR is to maintain or sacrifice minimal accuracy while providing items that offer novelty, diversity, and serendipity to users.

• **MOMDP Formulation of MOPSR.** We formulate MOPSR as a Multi-Objective Markov Decision Process (MOMDP) [36] and the Hierarchical RL (HRL) is introduced to solve the offline problem. The recommender is the agent, and the users are the environment. The agent (recommender) interacts with the environment (users), taking actions (recommending items) based on the state of the environment, and is updated by the feedback (reward) from the environment. Continually, the environment generates a new state for the agent as users respond to the actions. In HRL-based MOPSR, users provide inter-session and intra-session states, where the former guides the agent considering the dynamic preferences across sessions to predict items with the intra-session state.

• **Notation.** Our paper involves numerous notations. To facilitate clarity for future reading, we provide a reference Table 1 illustrating the important ones throughout this paper.

2.2 Overall Framework

Traditional offline RL-based recommender mimics the online process by observing the rewards w.r.t specific behaviors from users and is trained to maximize the expected cumulative rewards (expected returns) [36, 45]. However, this process becomes very complex when handling multi-objective tasks for offline data and HRL problems due to adding more agent/value heads [36, 55]. Following the Decision Transformer (DT) [3] that recast the offline RL as sequence modeling to predict actions in a supervised manner conditioned on the expected returns and states, we solve the HRL problem

Table 1: An illustration of notations.

	Description
I	The set of items
C_c	c -th session or the current session of a user
$x_{c,t}$	t -th user-item interaction in C_c , i.e., $x_{c,t} \in I$
s_c	Inter-session state session C_c
R_c	Inter-session unexpected return for session C_c
g_c	Goal state for session C_c
$\tilde{R}_c$	Goal return for session C_c .
$s_{c,t}$	Intra-session state at timestamp t in C_c
$R_{c,t}$	Intra-session unexpected return at timestamp t in C_c
$\tilde{R}_{c,t}^n$	Intra-session novelty return at timestamp t in C_c
$a_{c,t}$	Action at timestamp t in C_c , i.e., $a_{c,t} = x_{c,t+1}$

by the proposed Hierarchical Decision Transformers (HDT) framework as illustrated in Figure 1. HDT consists of two layers: 1) the Inter-session DT (InterDT) models dynamic long-term interest as the session evolves, and 2) the Intra-session DT (IntraDT) models short-term interests within the current session.

The MOPSR is naturally empowered by expected returns and user states at both inter- and intra-session levels. On the one hand, actions (items) are generated under the guidance of designed rewards and user states to balance accuracy with diversity, novelty, and serendipity objectives. On the other hand, to personalize the recommendation with users’ long-term interests, the goal state and reward of inter-session are conveyed to the intra-session state to jointly recommend the next items. Formally, the trajectory the HDT aims to model is:

$$\tau = (\dots, R_c, s_c, g_c, \tilde{R}_c, R_{c,1}^n, s_{c,1}, a_{c,1}, \dots, R_{c,T}^n, s_{c,T}, a_{c,T}, \dots) \quad (1)$$

The components of the trajectory are:

- Inter-session state space $\mathcal{S}^u$: $s_c \in \mathcal{S}^u$ is a user’s inter-session state at session C_c , providing representation of long-term preferences evolved across sessions. The initial state $s_1 \in \mathcal{S}^u$ for all users is initialized to zeros.

- Inter-session unexpected returns $\mathcal{R}^u$: $R_c \in \mathcal{R}^u$ represents the expected unexpectedness contributed by the interactions from session C_c to the final session for a user.
- Goal state space $\mathcal{G}$: $g_c \in \mathcal{G}$ is generated by the inter-session state s_c and the inter-session unexpected return R_c from the InterDT at session C_c . g_c will affect the intra-session recommendations by conveying a user's long-term interest in unexpected items to the intra-session state.
- Goal return space $\tilde{\mathcal{R}}^u$: $\tilde{R}_c \in \tilde{\mathcal{R}}^u$ is also generated by the inter-session state s_c and the inter-session unexpected return R_c from the InterDT at session C_c . Differently, $\tilde{R}_c$ is the output at the return position that will control the intra-session recommendations by producing the intra-session unexpected return.
- Intra-session state space $\mathcal{S}^z$: $s_{c,t} \in \mathcal{S}^z$ is a user's intra-session state at timestamp t within the session C_c , which represents the short-term interest of a user. $s_{c,t}$ is generated by a sequential model $\mathcal{E}$ that encodes the sequence $x_{c,1:t} = \{x_{c,1}, \dots, x_{c,t}\} \in C_c$ and the goal state g_c . When recommending items in session C_c , we also call it the current session C_c .
- Intra-session unexpected return $\mathcal{R}^z$: $R_{c,t} \in \mathcal{R}^z$ represents the expected unexpectedness achieved by the remaining interactions from timestamp t in session C_c , which reflects a user's short-term preference for unexpected items. Instead of comprising the input trajectory, $R_{c,t}$ is taken as the target unexpectedness in the training process to balance multiple objectives.
- Intra-session novelty return $\mathcal{R}^{nz}$: $R_{c,t}^n \in \mathcal{R}^{nz}$ denotes the novelty achieved by remaining interactions from step t in the session C_c , which reflects a user's short-term preference for novel items in the system.
- Action space $\mathcal{A}$: $\mathcal{A}$ is the discrete action set consisting of candidate items. In the offline setting, the action $a \in A$ at time step t in the session c is the next item, i.e., $a_{c,t} = x_{c,t+1}$.

We can see the overall training process from Figure 1: at the beginning of a session C_c , the InterDT observes the inter-session state s_c and the inter-session unexpected return R_c to produce the goal state g_c and goal return $\tilde{R}_c$ for intra-session recommendations. Then, the IntraDT incorporates the intra-session state $s_{c,t}$ (generated by user-item sequence $X_{c,1:t}$ and the goal state g_c), goal return $\tilde{R}_c$, and the intra-session novelty return $R_{c,t}^n$ to generate action (recommend item) a_t at timestamp t . Next, we detail designing returns and modeling trajectories in Inter-session and Intra-session DTs.

2.3 Inter-session Modeling

2.3.1 Inter-session Unexpected Return. To provide users with diverse recommendations that touch upon a broader selection of items as possible, we design an inter-session unexpected return R_c to quantify the maximum unexpectedness that future sessions will bring to the user starting from session C_c .

$$R_c = dis_{max} - dis(C_1, \dots, C_c) \quad (2)$$

where $dis(C_1, \dots, C_c)$ is the dissimilarity of previous c sessions, calculated as the average dissimilarity (1 minus average cosine similarity) among all interacted items' embeddings in the past c sessions. dis_{max} is the maximum dissimilarity the user can obtain by items in the system. R_c reflects users' propensity to explore diverse, novel, and unexpected items across sessions.

2.3.2 InterDT. Inter-session DT models R_c and s_c to generate the goal state and goal return for intra-session recommendation. R_c is mapped by a trainable embedding to form its representation with the same dimension as s_c . Then the inputs for InterDT at session C_c are formulated as:

$$H_c = stack(R_c, s_c) \in \mathbb{R}^{2 \times d}, \quad (3)$$

where d is the embedding size. The first state s_1 is initialized by zeros for each user. When session C_c ends, the s_{c+1} for the next session C_{c+1} is updated by the last intra-session state from session C_c , we illustrate the intra-session state in the following part. The goal return $\tilde{R}_c$ and goal state g_c are formulated as:

$$\begin{aligned} \tilde{H}_c &= T_1(H_c), \\ \tilde{R}_c, g_c &= unstack(\tilde{H}_c), \end{aligned} \quad (4)$$

where T_1 is a self-attention layer with residual connection [39].

2.4 Intra-session Modeling

2.4.1 Intra-session State. Given goal state g_c that represents long-term preference on diverse and unexpected items and the user-item interactions $x_{c,1:t}$ within the session C_c , the intra-session state $s_{c,t}$ is the hidden state from the sequential model [10, 15] G :

$$x_{c,t} = W_s[x_{c,t}, g_c] + b_s, \quad (5)$$

$$s_{c,t} = G(x_{c,1:t}), \quad (6)$$

2.4.2 Intra-session Novelty Returns. We introduce a novelty return $R_{c,t}^n$ to promote the recommendation of unknown/novel items [7, 36] that belongs to the less frequently interacted item set I_{nov} (within the bottom 90%) in the training data. In this way, recommending more unpopular items can lead to an increase in the system's diversity and novelty. In session C_c with T steps, $R_{c,t}^n$ is calculated as:

$$R_{c,t}^n = \# \text{ items } \in I_{nov} \text{ from step } t \text{ to } T \quad (7)$$

We expect to achieve maximum novelty and diversity, so the predicted item is treated as novel during inference to calculate the $R_{c,t}^n$, which is mapped by a trainable embedding to form its representation with the same dimension as $s_{c,t}$.

2.4.3 Intra-session Unexpected Return. $R_{c,t}$ aims to represent the unexpectedness achieved by the remaining interactions from timestamp t to the end of session C_c :

$$R_{c,t} = dis(x_{c,1}, \dots, x_{c,T}) - dis(x_{c,1}, \dots, x_{c,t}) \quad (8)$$

where T is the length of session c . $dis(x_{c,1}, \dots, x_{c,t})$ is the average dissimilarity (1 minus average cosine similarity) of previous ts interacted items' embeddings in session c . During the training phase, we treat $R_{c,t}$ as the prediction target rather than an input, with the aim of eliminating the need for searching for optimal settings during the inference stage.

2.4.4 IntraDT. Given the goal return $\tilde{R}_c$, intra-session novelty return $R_{c,t}^n$, intra-session state $s_{c,t}$, and action a_t (next item), intra-session DT models $(\tilde{R}_c, R_{c,t}^n, s_{c,t}, a_{c,t})$ sequentially. Specifically, the input at timestamp t is constructed as:

$$Z_{c,t} = stack(\tilde{R}_c, R_{c,t}^n, s_{c,t}, a_{c,t}) \in \mathbb{R}^{4 \times d}, \quad (9)$$

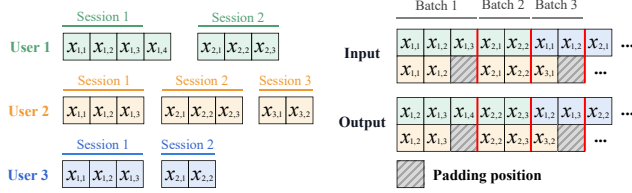


Figure 2: Left: user data. Right: user-parallel mini-batches for batch size 2.

Therefore, the complete input sequence of Z_c for InterDT is:

$$Z_c = \{Z_{c,1}, \dots, Z_{c,T}\}, \quad (10)$$

$$\tilde{Z}_c = T_2(Z_c), \quad (11)$$

where $\tilde{Z} = \{\tilde{Z}_{c,1}, \dots, \tilde{Z}_{c,T}\}$. T_2 is the GPT [30] architecture to autoregressively predict actions conditioned on returns and states. The hidden state $\tilde{R}_{c,t}$ and $\tilde{s}_{c,t}$ at each step t are then unstacked from $\tilde{Z}_t$:

$$\tilde{R}_{c,t}, \tilde{s}_{c,t} = \text{unstack}(\tilde{Z}_{c,t}), \quad (12)$$

where $\tilde{R}_{c,t}$ is mapped by an MLP layer to produce the predicted intra-session unexpected return $\hat{R}_{c,t}$:

$$\hat{R}_{c,t} = W_r \tilde{R}_{c,t} + b_r, \quad (13)$$

where $W_r \in \mathbb{R}^{d \times 1}$ are trainable parameters, $b_r \in \mathbb{R}^1$ are bias vectors. The actions (next items) are subsequently predicted based on the hidden state $\tilde{s}_{c,t}$:

$$\hat{a}_{c,i} = \text{softmax}(\delta(W_a \tilde{s}_{c,t} + b_a)), \quad (14)$$

where δ denotes the activation function, $\hat{a}_{c,i}$ is the predicted score on i -th candidate action, $W_a \in \mathbb{R}^{d \times N}$ are trainable parameters, and $b_a \in \mathbb{R}^N$ are bias vectors.

2.5 Overall Learning

2.5.1 Training Objective. The overall learning optimization process of HDT consists of a goal loss in InterDT, a sequential loss, a return loss, and an action loss in IntraDT.

In InterDT, g_c is used to predict the first action (item) in the current session C_c via the goal loss:

$$\hat{y}_i = \text{softmax}(\delta(W_u g_c + b_u)), \quad (15)$$

$$\mathcal{L}_g = -\sum_{i=1}^n y_i \log(\hat{y}_i). \quad (16)$$

where δ is the activation function, $W_u \in \mathbb{R}^{d \times N}$ are trainable parameters, $b_u \in \mathbb{R}^N$ is the bias vector. $\hat{y}_i$ is the probability that the user interacted with i -th item at the first step of session C_c . $y_i = 1$ if the session starts with the i -th item, otherwise $y_i = 0$.

The sequential model G is optimized by a sequential loss to predict next items in a session:

$$\hat{y}_{c,i} = \text{softmax}(\delta(W_s s_{c,t} + b_s)), \quad (17)$$

$$\mathcal{L}_S = -\sum_{i=1}^n y_{c,i} \log(\hat{y}_{c,i}). \quad (18)$$

where $s_{c,t}$ is mapped into n classification logits on the candidate items by a fully connected layer with softmax function. $W_s \in \mathbb{R}^{d \times N}$ are trainable parameters, and $b_s \in \mathbb{R}^N$ is the bias vector. $y_{c,i}$ is the logit of i -th action. $W_s \in \mathbb{R}^{d \times N}$ are trainable parameters, and $b_s \in \mathbb{R}^N$ is the bias vector. $y_{c,i} = 1$ if the user interacted with the

i -th item at the next time step in the current session c . Otherwise, $y_{c,i} = 0$.

In IntraDT, to enhance the stability of the balance between accuracy and diversity, novelty, and serendipity objectives, we also introduce the learning of the intra-session unexpected return by the Mean Squared Error (MSE) loss:

$$\mathcal{L}_R = (1/T) \sum_{t=1}^T (R_{c,t} - \hat{R}_{c,t})^2. \quad (19)$$

where $R_{c,t}$ is the target intra-session unexpected return, $\hat{R}_{c,t}$ is the predicted value.

Furthermore, an action loss in a cross-entropy manner is used to predict the actions (next items):

$$\mathcal{L}_a = -(1/T) \sum_{t=1}^T \sum_{i=1}^n a_{c,t} \log(\hat{a}_{c,i}). \quad (20)$$

$a_{c,i} = 1$ if the user interacted with the i -th item at the next time step in session c . Otherwise, $a_{c,i} = 0$.

Finally, we jointly train all the loss on the training data to update all components of HDT:

$$\mathcal{L} = \mathcal{L}_S + \alpha \mathcal{L}_R + \beta \mathcal{L}_a + \lambda \mathcal{L}_g \quad (21)$$

where α and β are hyperparameters that control the weights of intra-session unexpected return prediction loss and intra-session action prediction loss. λ is the weight of the first action prediction loss in the inter-session layer. In the inference, items are ranked by the scores from the sequential model and the interDT. The impact of these hyperparameters will be discussed in section 3.6.

2.5.2 Efficient Offline Training. Algorithm 1 outlines the procedure for generating the offline training set $\mathcal{D}_{tr}$ from training data $\mathcal{D}_{ts}$. Specifically, the HDT framework is trained in an end-to-end manner, while the InterDT is updated between sessions. The InterDT captures the evolution of user preferences across sessions sequentially, which personalizes the sequential model and IntraDT then generates recommendations. To improve training efficiency, we adapt the user-parallel mini-batch training strategy [29] to HDT, where sessions are grouped by users and sorted by timestamp (see Figure 2). In the iteration, the initial session with sequence $\{x_{1,1}, \dots, x_{1,T}\}$ of the first B users are fed into HDT; the

Algorithm 1 Offline training set generation.

Input: user-item interaction sessions set $\mathcal{D}_{ts}$, novel item set I_{nov}

Output: multi-objective training set $\mathcal{D}_{tr}$

```

1: repeat
2:   Sample a user with sequential sessions  $C = \{C_1, \dots, C_m\}$ 
3:   from  $\mathcal{D}_{ts}$ , where  $C_c = \{x_{c,1}, \dots, x_{c,T}\}, c = 1, \dots, m$ 
4:   for  $c=1 : m$  do
5:     Compute  $R_c$  according to Eq. 2
6:     for  $i=1 : T$  do
7:       Compute  $R_{c,1:T}$  according to Eq. 8
8:       Compute  $R_{c,1:T}^n$  according to Eq. 7
9:     end for
10:   end for
11:    $\mathcal{D}_{tr}.append(C, R_{1:m}, R_{1:m,1:T}, R_{1:m,1:T}^n)$ 
12:    $\mathcal{D}_{ts}.remove(C)$ 
13: until  $\mathcal{D}_{ts} = \emptyset$ 
14: return  $\mathcal{D}_{tr}$ 

```

Table 2: Dataset statistics († denotes mean).

Dataset	Album	Reddit
Users	14,341	11,160
Items	129,273	21,583
Sessions	115,225	321,878
Events	1,051,698	1,234,442
Events per item †	8.13	57.19
Events per session †	9.12	3.83
Sessions per user †	8.03	28.84
Training - Events	915,195	1,194,739
Training - Sessions	100,890	310,719
Test - Events	136,503	39,703
Test - Sessions	14,335	11,159

sequence $\{x_{1,2}, \dots, x_{1,T+1}\}$ is used to construct the sequence of next items/actions for the intra-session part. In the next iteration, the second session of the first B users continue the process, and so on. Sessions of users are processed in chronological order. Once a user has been completely processed, the inter-session state s_1 to InterDT is reset and the next new user takes its place in the mini-batch. With both generative training and user-parallel mini-batches, we can train HDT efficiently over users with different numbers of sessions.

3 EXPERIMENTS

In this paper, we aim to address the following research questions:

RQ1: How does HDT architecture personalize the sequential methods to improve user satisfaction across sessions, focusing on the objectives of diversity, novelty, and serendipity without sacrificing the accuracy performance?

RQ2: How do the intra-session unexpected and novelty returns affect the recommendation in current session?

RQ3: How to select weights of training objectives to balance accuracy, diversity, novelty, and serendipity?

3.1 Datasets

We perform experiments on two real-world datasets: **Album** [34] and **Reddit** [29]. The detailed statistics are summarized in Table 2. The Album dataset contains one-week listening events of users on albums, collected by the music streaming platform Last.fm¹. The second Reddit dataset came from the Reddit social media platform and includes interactions made by users over a five-day period.

To reduce the noise in the Album dataset, we remove consecutive repeated interactions while retaining the first timestamp. Both datasets are preprocessed following the approach described in [29], and user interactions are divided into sessions using a 60-minute threshold. More precisely, we filter items with less than three interactions. Then, sessions with less than three interactions are excluded. Finally, users with at least five sessions are retained to ensure sufficient cross-session information. For hyperparameter tuning, we further partition the training set using the same procedure to create a validation dataset. We conduct each experiment² five times and report the average results.

¹<http://www.last.fm/api>

²https://drive.google.com/drive/folders/1AnN7dyf3_-i79ZPfvQe-cSjJlqcujuvBU?usp=sharing

Same as [29], the testing set consists of all interactions of the last session of each user, and the sessions before the test (last) sessions form the training set. For example, for a user with sessions $C = \{C_1, C_2, C_3\}$, the training set are the $C_1 = \{x_{1,1}, x_{1,2}\}$ and $C_2 = \{x_{2,1}, x_{2,2}, x_{2,3}\}$, and the last session $C_3 = \{x_{3,1}, x_{3,2}, x_{3,3}\}$ is the testing set. During the inference stage, historical sessions, current session, and truth next item are built as $\{\{x_{1,1}, x_{1,2}\}, \{x_{2,1}, x_{2,2}, x_{2,3}\}\}$, $\{x_{3,1}\}$, and $x_{3,2}$; $\{\{x_{1,1}, x_{1,2}\}, \{x_{2,1}, x_{2,2}, x_{2,3}\}\}$, $\{x_{3,1}, x_{3,2}\}$, and $x_{3,3}$.

3.2 Baselines and Settings

Our proposed HDT can personalize many sequential models to recommend the next items by strictly adhering to the sequential dependencies across sessions and order information within a recent session. To verify its generalization, we integrate HDT with four well-known sequential recommendation models: •GRU4Rec [10] that employs a GRU layer to capture user interactions within a session. •Caser [38] that utilizes convolution operations to process previous item embeddings based on CNN. •NItNet [52] that improves the receptive field via dilated CNN operations and deepens the network by residual connections. •SASRec [15] that is based on self-attention to encode sequential items in a session.

Each sequential recommendation model is adapted and compared with the following approaches:

- Normal: Sequential recommendation models Trained by the normal cross-entropy loss. Sessions from the same user are concatenated into a single session to form the training data.
- SMORL [36]: A state-of-the-art RL-based multi-objective sequential recommendation model that balances accuracy, diversity, and novelty based on Q-learning to reward each interaction. We implement it with the same setting as the Normal method.
- Hier [29, 50]: The only hierarchical structure-based PSR models that recommend items following strictly order information across and within sessions. Besides the original RNN [29] and CNN-based [50] methods, we extend Hier to SASRec for comprehensive comparison.
- InterDT: Hier methods that incorporate inter-session DT. Item prediction is done by the intra-session state and goal state.
- HDT: Our proposed Hierarchical Decision Transformers.

The length of user sessions is limited to 20, and longer sessions are truncated to retain only the most recent interactions. Padding tokens are added to shorter sessions in the end to ensure the same length. The user mini-batch size is set to 100 for the Album dataset and 200 for the Reddit dataset. To ensure a consistent and fair comparison, all models and variations are trained using the Adam optimizer [36], and the learning rate is searched from [0.005, 0.001, 0.0005, 0.0001]. Evaluation of the validation set is performed every 1,000 batches of updates. The embedding sizes for item, state, return, and action are all set to 64. For GRU, the hidden state size is also set to 64. Caser uses 1 vertical convolution filter and 16 horizontal filters with heights set from 2,3,4. NItNet follows the parameters described in the original paper. For SASRec, the number of heads in self-attention is set to 1, and the dropout ratio is set to 0.1, as described in [15].

Table 3: Results on Album dataset. NG denotes NDCG. Unexp. denotes Unexpectedness. Boldface denotes highest scores. Second-best scores are marked with . * denotes the significance p -value < 0.05 compared to best baseline of first three models.

Models	Accuracy				Diversity		Novelty		Unexp.		Serendipity	
	HR@10	NG@10	HR@20	NG@20	CV _d @5	CV _d @10	CV _n @5	CV _n @10	UP@5	UP@10	HR _{ser} @10	NG _{ser} @10
GRU4Rec	0.0761	0.0485	0.0994	0.0543	0.4151	0.5285	0.3551	0.4701	2.524	3.462	0.0678	0.0463
GRU-SMORL	0.0765	0.0484	0.0983	0.0563	0.4739	0.5909	0.4102	0.5472	2.932	4.202	0.0842	0.0573
Hier-GRU	0.0814	0.0568	0.1004	0.0622	0.4119	0.5199	0.3602	0.4799	2.885	4.112	0.0743	0.0469
InterDT-GRU	<u>0.0845</u>	<u>0.0574</u>	<u>0.1086</u>	<u>0.0629</u>	<u>0.5411</u>	<u>0.6601</u>	<u>0.4691</u>	<u>0.6069</u>	<u>3.034</u>	<u>4.216</u>	<u>0.0862</u>	<u>0.0591</u>
HDT-GRU	0.0896*	0.0628*	0.1107*	0.0681*	0.5846*	0.6948*	0.5395*	0.6611*	3.122*	4.301*	0.1023*	0.0769*
Caser	0.0637	0.0291	0.0932	0.0365	0.2617	0.3428	0.1906	0.2752	2.811	3.826	0.0655	0.0337
Caser-SMORL	0.0643	0.0299	0.0968	0.0385	0.2844	0.3758	0.2356	0.3245	2.967	3.967	0.0795	0.0519
Hier-Caser	0.0634	0.0298	0.0945	0.0376	0.2301	0.2984	0.1567	0.2253	2.535	3.531	0.0764	0.0507
InterDT-Caser	<u>0.0661</u>	<u>0.0313</u>	<u>0.0972</u>	<u>0.0392</u>	0.2748	0.3632	0.2305	0.3199	2.667	3.713	0.0792	0.0511
HDT-Caser	0.0783*	0.0435*	0.1084*	0.0453*	0.3024*	0.4122*	0.2458*	0.3515*	3.103*	4.355*	0.0843*	0.0553*
NitNet	0.0803	0.0548	0.1007	0.0603	0.5402	0.6468	0.5234	0.6424	2.955	3.873	0.1005	0.0632
NitNet-SMORL	0.0813	0.0552	0.1009	0.0612	0.5834	0.6956	0.5432	0.6601	3.462	4.474	0.1019	0.0628
Hier-NitNet	0.0862	0.0614	0.1041	0.0658	0.5515	0.6757	0.5081	0.6427	3.314	4.366	0.1093	0.0711
InterDT-NitNet	<u>0.0920</u>	<u>0.0666</u>	<u>0.1091</u>	<u>0.0708</u>	<u>0.5936</u>	<u>0.7126</u>	<u>0.5546</u>	<u>0.6856</u>	<u>3.514</u>	<u>4.576</u>	<u>0.1181</u>	<u>0.0799</u>
HDT-NitNet	0.0946*	0.0673*	0.1127*	0.0719*	0.6241*	0.7368*	0.5837*	0.7078*	3.622*	4.782*	0.1219*	0.0809*
SASRec	0.1061	0.0742	0.1282	0.0798	0.4728	0.5743	0.4186	0.5291	3.495	4.563	0.1292	0.0813
SAS-SMORL	0.1084	0.0753	0.1301	0.0808	0.5144	0.6186	0.4642	0.5914	4.092	5.364	0.1486	0.0912
Hier-SAS	0.1163	0.0786	0.1431	0.0853	0.4809	0.5851	0.4276	0.5411	3.957	5.253	0.1445	0.0882
InterDT-SAS	<u>0.1178</u>	<u>0.0802</u>	<u>0.1451</u>	<u>0.0871</u>	0.4923	0.5951	0.4152	0.5476	4.014	5.327	0.1467	0.0903
HDT-SAS	0.1202*	0.0816*	0.1462*	0.0879*	0.5742*	0.6891*	0.5285*	0.6548*	4.195*	5.525*	0.1503*	0.0924*

3.3 Metrics

We apply the following metrics to comprehensively evaluate the performance of HDT in balancing multiple objectives.

Accuracy. Hit Ratio (HR@ k) [36] and Normalized Discounted Cumulative Gain (NDCG@ k) [13] are applied to measure the relevance of recommended items, where $k \in \{10, 20\}$. We report the average results over all interactions of the test sessions.

Diversity & Novelty. We measure the aggregate diversity and novelty at the system level by the commonly applied Coverage (CV) [14] metrics. CV_d@ k , $k \in \{5, 10\}$ for diversity is calculated as a percentage of the whole item set covered by all top- k recommended items for all events in test sessions. CV_n@ k for novelty is calculated based on the novel items.

Unexpectedness & Serendipity. Unexpectedness (UP@ k) metric, where $k \in \{5, 10\}$, evaluates the ability of an RS to recommend items that are unexpected for a user given his historical sessions. In our paper, UP@ k for a user is formulated as the disparity between the top- k recommended items for current session and the interacted items in the past sessions, i.e., $UP@k = \frac{1}{U} \sum_{u=1}^U UNSEEN_u$, where $UNSEEN_u$ is the count of items in the top- k recommendation list that are not interacted by user u before. The unexpectedness is reported as the average value of all users in the test set.

To evaluate serendipity, i.e., unexpected and relevant items for a user, we adopt HR_{ser}@ k [8], where $k \in \{10\}$, to measure if a serendipity item (ground truth and unexpected item for the user) is recommended in top- k list. Similar to accuracy metrics, the Serendipity-Based Normalized Discounted Cumulative Gain (NDCG_{ser}@ k) [8] is also used to calculate the rank of the serendipity items. The results are reported as the average value over all serendipity interactions of the test sessions.

3.4 Performance Comparison (RQ1)

In this section, we analyze the performance of various recommendation methods across multiple objectives using the Album and Reddit datasets, as detailed in Tables 3 and 4. Our HD method outperforms other techniques, achieving the best balance across all evaluated metrics. For instance, when employing the Caser-based approaches on the Album dataset, HDT-Caser enhances performance by 18.4% in HR@10, 13.4% in CV_d@10, 17.2% in UP@10, and 6.4% in HR_{ser}@10, all relative to InterDT-Caser. HDT not only sets new benchmarks in these metrics but also uniquely improves both accuracy and unexpectedness, a combination where conventional sequential methods typically show high unexpectedness but lower accuracy. While GRU-SMORL and Caser-SMORL on the Reddit dataset achieve marginally better results in diversity and novelty, they fall short of HDT’s performance across other metrics.

Additionally, Hierarchical (Hier) method exhibits strong improvements in accuracy-related metrics; for example, Hier-NitNet on the Album dataset reports up to a 12% increase in NDCG@10. This underscores the significant influence of modeling the evolution of user preferences across sessions. However, it is notable that the improvements in accuracy by Hier- models do not extend equally to diversity and serendipity. For instance, while Hier-GRU on the Album dataset shows a 6.96% improvement in HR@10 over standard GRU, it suffers from noticeable declines in diversity and novelty. This trend is consistent across other hierarchical implementations like Hier-Caser and Hier-NitNet on the Reddit dataset when compared with baseline methods like Caser and NitNet.

InterDT models are more effective compared with Hier methods, particularly InterDT-GRU and InterDT-NitNet, which enhance accuracy, diversity, and serendipity concurrently.

Table 4: Results on Reddit dataset. NG denotes NDCG. Unexp. denotes Unexpectedness. Boldface denotes highest scores. Second-best scores are marked with . * denotes the significance p -value < 0.05 compared to best baseline of first three models.

Models	Accuracy				Diversity		Novelty		Unexp.		Serendipity	
	HR@10	NG@10	HR@20	NG@20	CV _d @5	CV _d @10	CV _n @5	CV _n @10	UP@5	UP@10	HR _{ser} @10	NG _{ser} @10
GRU4Rec	0.2653	0.1615	0.3448	0.1816	0.3695	0.4843	0.3042	0.4278	3.058	4.755	0.2855	0.1821
GRU-SMORL	0.2948	0.1892	0.3858	0.2013	0.3954	0.5045	0.3423	0.4544	3.145	4.996	0.3171	0.2252
Hier-GRU	0.3711	0.2205	0.4769	0.2473	0.2778	0.3673	0.2161	0.3069	3.099	5.037	0.3052	0.2131
InterDT-GRU	0.3675	0.2156	0.4703	0.2484	0.2949	0.3783	0.2302	0.3171	3.181	5.121	0.3101	0.2169
HDT-GRU	0.3754	0.2245	0.4829*	0.2517*	0.3869	0.4871	0.3581*	0.4454	3.257*	5.186*	0.3361*	0.2355*
Caser	0.2489	0.1165	0.3482	0.1416	0.2001	0.2344	0.1143	0.1556	2.882	4.754	0.2391	0.1092
Caser-SMORL	0.2632	0.1376	0.3685	0.1725	0.2265	0.2644	0.1378	0.1694	3.240	5.145	0.2412	0.1122
Hier-Caser	0.3159	0.1699	0.4192	0.1961	0.1638	0.1957	0.0798	0.1088	3.845	5.703	0.2377	0.1117
NitDT-Caser	0.3109	0.1723	0.4082	0.1998	0.1663	0.1968	0.0873	0.1149	3.878	5.722	0.2481	0.1177
HDT-Caser	0.3212*	0.1745*	0.4239*	0.2005*	0.2067	0.2541	0.1368	0.1796*	3.928*	5.837*	0.2937	0.1632*
NitNet	0.2421	0.1437	0.3235	0.1643	0.2496	0.3193	0.1858	0.2531	2.961	4.696	0.2612	0.1525
NitNet-SMORL	0.2741	0.1767	0.3623	0.1842	0.2606	0.3264	0.2081	0.2734	3.143	4.991	0.3047	0.1875
Hier-NitNet	0.3144	0.1845	0.4135	0.2096	0.2112	0.2793	0.1495	0.2128	3.132	4.926	0.3022	0.1793
InterDT-NitNet	0.3215	0.1917	0.4185	0.2162	0.2001	0.2608	0.1312	0.1879	3.214	5.066	0.3149	0.1997
HDT-NitNet	0.3368*	0.2072*	0.4311*	0.2312*	0.2849*	0.3599*	0.2231*	0.2976*	3.443*	5.288*	0.3547*	0.2204*
SASRec	0.2667	0.1504	0.3692	0.1743	0.2314	0.2869	0.2098	0.2571	2.861	4.542	0.2765	0.1552
SASRec-SMORL	0.2845	0.1856	0.3923	0.2054	0.2954	0.3867	0.2654	0.3462	3.634	5.584	0.3362	0.2363
Hier-SASRec	0.3937	0.2342	0.4969	0.2603	0.2891	0.3677	0.2475	0.3254	3.381	5.361	0.3854	0.2175
InterDT-SASRec	0.4032	0.2519	0.5017	0.2767	0.2681	0.3366	0.2278	0.2884	3.721	5.736	0.4182	0.2603
HDT-SASRec	0.4073*	0.2544*	0.5092*	0.2802*	0.3445*	0.4331*	0.2972*	0.3841*	3.744*	5.755*	0.4354*	0.2711*

Table 5: Effect of intra-session returns on Accuracy, Diversity, Novelty, Unexpectedness, Serendipity. uxp/nov denotes the variants of integrating either the intra-session unexpected or novelty return. Boldface denotes best results.

Methods		Acc.	Div.	Nov.	UP.	Ser.
		HR@10	CV _d @5	CV _n @5	UP@5	HR _{ser} @10
Album	uxp	0.1182	0.5272	0.4768	4.042	0.1482
	nov	0.1197	0.5706	0.5264	4.001	0.1242
	HDT	0.1202	0.5742	0.5285	4.195	0.1503
Reddit	uxp	0.4036	0.3209	0.2575	3.721	0.4123
	nov	0.4021	0.3408	0.2998	3.311	0.4054
	HDT	0.4073	0.3445	0.2972	3.744	0.4354

Specifically, InterDT-Caser on the Album dataset shows increases of about 4.2% in HR@10, 41.9% in CV_d@10, 5.2% in UP@10, and 3.6% in HR_{ser}@10. These gains are attributed to the model’s capability to track the user’s goal state across multiple objectives effectively.

In conclusion, while hierarchical models excel in capturing user interest evolution to recommend more accurate items, they do not equally enhance other objectives like diversity and serendipity. While HDT achieves a superior balance across all metrics, setting a new standard for multi-objective recommendation systems.

3.5 Intra-session Return Investigation (RQ2)

We examine the impact of intra-session unexpected (uxp) and novelty (nov) returns by the IntraDT-SAS model, analyzing their influence on the performance of Album dataset. The results are presented in Table 5. Considering both nov and uxp, we note that they achieve comparable accuracy performance, with uxp achieving higher scores on serendipity and unexpectedness, while nov

performs higher in diversity and novelty. This demonstrates that the intra-session unexpectedness return guides the model to recommend items that the user has not interacted with personally, while the novelty return encourages recommending unexplored/new items of the system. Moreover, HDT consistently outperforms other variants across all evaluated objectives, confirming its robustness in effectively integrating multiple objective returns to enhance the recommendation quality. On the Reddit, the performance of uxp and nov appears to be comparable, further underscoring the versatility of the HDT in different contexts. The findings highlight the utility of tailored intra-session returns in guiding the recommendation process toward achieving specific objectives and the superior capability of HDT to deliver multiple benefits.

3.6 Hyperparameter Study (RQ3)

We optimize three key hyperparameters: α , β , and λ , which control the weights of the return loss $\mathcal{L}_R$, action loss $\mathcal{L}_a$, and goal loss $\mathcal{L}_g$ in Eq.21, respectively, influencing the model’s ability to balance between accuracy and other performance metrics such as unexpectedness and diversity. We optimize one weight at a time and individually observe the changes in accuracy and the corresponding objective metric to determine the optimal value that balances the two objectives. Due to space constraints, we present the results for the Reddit dataset on a GRU-based model.

Figure 3a illustrates the impact of α , which controls the influence of the return loss $\mathcal{L}_R$, on the metrics HR@20 and UP@10. We observe that both metrics improve as α increases from 0 to 0.2. While HR@20 begins to decline when α continues to increase beyond 0.2, UP@10 continues to show positive growth. Based on these trends, we set α to 0.5 to optimize HR@20 and UP@10 simultaneously.

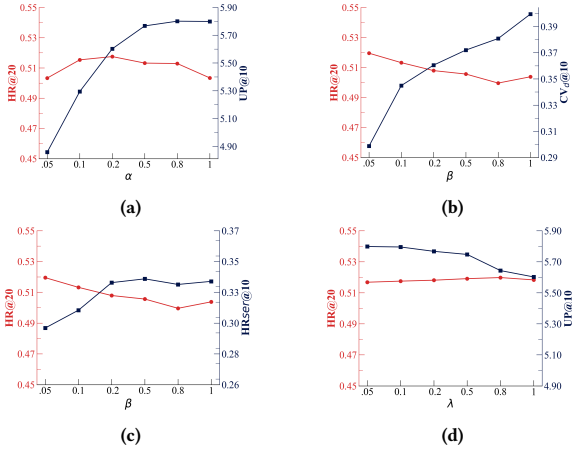


Figure 3: HDT with different weights of distinct objective loss on the Reddit dataset.

Diversity and novelty are mainly affected by the action loss $\mathcal{L}_a$. Figure 3b shows that increasing β results in a steady increase in CV (we show $CV_d@10$) and a slow decrease in HR@20. We set β as 0.2 to maintain a reasonable balance between diversity, novelty and accuracy. We observe influence of λ on HR@20 and UP@10 by ensuring accuracy while reducing potential adverse effects of goal loss $\mathcal{L}_g$ on other objectives. We set λ to 0.1 which best supports accuracy without excessively compromising unexpectedness.

For the Album dataset, the optimal settings for α , β , and λ are found to be 0.5, 0.1, and 0.2 respectively. These values reflect a tailored approach to balancing the multiple objectives of our recommendation system, ensuring that accuracy is maintained while also enhancing other vital metrics like diversity, novelty and serendipity.

4 RELATED WORK

Personalized Session-based Recommendation. PSR was first proposed by Quadrana et al. [29] to recommend next item given strict order information within current session. Different from traditional sequential or session-based recommendation [10, 15, 38, 41] that learns ordered/unordered information within a recent session, PSR typically relies on both sequential dependencies inside sessions (intra-session) and across sessions (inter-session). A few studies [29, 50] have attempted to integrate incrementally increased sequential information for PSR by long-term preferences from historical sessions and short-term interests in the current sessions. Quadrana et al. [29] firstly personalized RNNs to model users’ historical sessions over time via a hierarchical structure (HRNN). You et al. [50] follows HRNN and adapts the hierarchical structure to CNN-based recommender. Although some GNN-based [27, 49, 53] methods proposed to represent items/users by pre-extracted global/local graphs from offline static data for PSR, these methods cannot learn users’ dynamic preferences in a recommendation scenario where user session comes sequentially. They fall outside the scope of our research. To be practical, we focus on improving the hierarchical method Quadrana et al. [29] that models the user’s historical sessions and current interactions both strictly in order.

Diversity and Novelty for Recommendation. Many other RSs have proposed multi-objective approaches, which focused on

pre-processing [16, 19, 22] or post-processing techniques [1, 2, 5, 12, 28, 35] for diversity, novelty or serendipity. In deep learning-based methods, research [17, 23, 32, 42, 44] investigated how to improve multi-objective during model training. Lee [17], Li et al. [23] focused on learning a representation of diversity/novelty to denote users’ preference and then introduced a learning module, e.g., attention [32, 42, 44], Transformer [8], to capture diverse patterns from historical data. Other research [4, 20, 25, 48, 54, 60] have attempted to balance accuracy with novelty and serendipity, by incorporating both accuracy and unexpectedness/novelty components into the training of a joint loss function. Pareto-based approaches [24, 26, 33] have also been suggested for optimizing multiple objectives. Rather than maximizing a single objective, these methods generate a range of solutions where no one solution is unequivocally the best across all objectives. However, there is no PSR that systematically sets baselines and evaluates accuracy with diversity, novelty, and serendipity.

Deep Reinforcement Learning for Session-based Recommendation. RL-based recommendation [36, 40, 45] is further introduced to train the RL agent to promote items. To improve diversity, Zheng et al. [56] proposed an exploration-exploitation strategy that selects random items near the recommended item. Although limited PSR methods are based on RL, we can find related work in a similar area of session-based recommendation. Xin et al. [47] introduced Prompt-Based Reinforcement Learning (PRL) to separably improve accuracy on distinct behaviours. Recent studies [9, 21, 36] formulated multi-objective sequential recommendations as RL problems to balance multiple objectives. Hansen et al. [9] introduced a simple ranker to provide items that the learned model might not generate. A supervised multi-objective reinforcement learning (SMORL) method is proposed to balance accuracy, diversity and novelty. The items are rewarded corresponding to each objective and the individual RL agent is trained to maximize the expected objective returns. Zhou et al. [57] extends SMORL with negative sampling to improve the multi-objective metrics. While having overlapping objectives with our method, they do not consider the personalized serendipity and long-term interest obtained from past sessions.

5 CONCLUSION AND FUTURE WORK

We proposed the novel Hierarchical Decision Transformers (HDT) framework for Personalized Session-based Recommendation (PSR). This framework models users’ long-term interest changes across past sessions and their short-term preferences within the current session. To enhance user experience by addressing multiple objectives, we introduced the Multi-objective PSR (MOPSR) and integrated both inter-session and intra-session unexpected returns into the HDT. This integration aims to guide recommendations by balancing accuracy with diversity, novelty, and serendipity.

We established specific metrics for each objective and conducted comprehensive experiments on publicly available datasets. The results from these comparisons underscored the effectiveness of our method in managing the trade-offs between conflicting objectives in MOPSR. In the future, we plan to extend the proposed framework to encompass additional recommendation scenarios and balance more objectives, thereby broadening its applicability in real world.

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